

Bessel's Differential eqn.

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0 \quad \longrightarrow \textcircled{1}$$

Eqn. (1) is known as Bessel's function.

Where n is an integer or half integer.

Eqn (1) can be written as

$$\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + \left(1 - \frac{n^2}{x^2}\right)y = 0 \quad \longrightarrow \textcircled{2}$$

Since $x=0$ is regular singular point for (2), so let its series

Solution: $y = \sum_{l=0}^{\infty} a_l x^{k+l} \quad \longrightarrow \textcircled{2}$

on Differentiating

$$\frac{dy}{dx} = \sum_{l=0}^{\infty} a_l (k+l-1) x^{k+l-1}$$

$$\frac{d^2 y}{dx^2} = \sum_{l=0}^{\infty} a_l (k+l-1)(k+l-2) x^{k+l-2}$$

Put values of $y, \frac{dy}{dx}, \frac{d^2 y}{dx^2}$
in eqn. (1)

$$\Rightarrow \sum_{l=0}^{\infty} a_l (k+l-1)(k+l) x^{k+l-2} + \frac{1}{x} \sum_{l=0}^{\infty} a_l (k+l) x^{k+l-1} + \left(1 - \frac{n^2}{x^2}\right) \sum_{l=0}^{\infty} a_l x^{k+l} = 0$$

$$\Rightarrow \sum_{l=0}^{\infty} a_l \left[(k+l-1)(k+l) x^{k+l-2} + (k+l) x^{k+l-2} + x^{k+l} - n^2 x^{k+l-2} \right] = 0$$

$$\Rightarrow \sum_{l=0}^{\infty} a_l \left[\left\{ (k+l-1)(k+l) + (k+l) - n^2 \right\} x^{k+l-2} + x^{k+l} \right] = 0$$

$$\boxed{\begin{aligned} & (k+l-1)(k+l) + (k+l) \\ &= (k+l) [k+l-1+1] = (k+l)^2 \end{aligned}}$$

$$\Rightarrow \sum_{l=0}^{\infty} a_l \left[\left\{ (k+l)^2 - n^2 \right\} x^{k+l-2} + x^{k+l} \right] = 0$$

$$\Rightarrow \sum_{l=0}^{\infty} a_l \left[(k+l+n)(k+l-n) x^{k+l-2} + x^{k+l} \right] = 0$$

Here for $l=0$, the smallest power of x is x^{k-2} , Equating its

Coefficient to zero for indicial eqn.

$$a_0 (k+n) (k-n) x^{k-2} = 0$$

$$k^2 - n^2 = 0$$

$$\begin{pmatrix} a_0 \neq 0 \\ x \neq 0 \end{pmatrix}$$

$$\boxed{k = \pm n}$$

Part 1 = 1 Equate zero of coefficient of x^{k-1}

$$a_1 (k+1+n) (k+1-n) x^{k-1} = 0$$

$$a_1 ((k+1)^2 - n^2) = 0 \quad \because x \neq 0$$

$$\Rightarrow \boxed{a_1 = 0} \text{ or } (k+1)^2 - n^2 = 0 \quad (*)$$

NOT possible because
for $k = \pm n$ eqn (*)
is NOT satisfied.

Now for recurrence formula, equate zero
of coefficient of x^{k+1} .

③ $\Rightarrow$ replace z by $z+2$.

$$a_{k+2} (k+1+2+n) (k+1+2-n) x^{k+2} + a_k x^{k+2} = 0$$

$$a_{l+2} (k+l+2+n)(k+l+2-n) + a_l = 0$$

$$\because k+n \neq 0$$

$$a_{l+2} = - \frac{a_l}{(k+l+2+n)(k+l+2-n)}$$

Here for $l = 1, 3, 5, 7, \dots$ 7

$$a_3 = - \frac{a_1}{(k+3+n)(k+3-n)} \Rightarrow a_5 = - \frac{a_3}{(k+5+n)(k+5-n)}$$

$$a_1 = a_3 = a_5 = \dots = 0$$

Now Two Cases arises.

(i) when $k=n$

$$a_{l+2} = - \frac{a_l}{(2n+l+2)(l+2)}$$

Put $l = 0, 2, 4, \dots$

$$a_2 = - \frac{a_0}{(2n+2) \cdot 2} = - \frac{a_0}{2^2(n+1)}$$

$$a_4 = \frac{-a_2}{(2n+4) \cdot 4} = \frac{-a_2}{2^3 (n+2)}$$

$$= \frac{-1}{2^3 (n+2)} \left(\frac{-a_0}{2^2 (n+1)} \right)$$

$$\boxed{a_4 = \frac{a_0}{2^4 2! (n+1)(n+2)}}$$

$$a_6 = \frac{-a_4}{(2n+6) \cdot 6} = \frac{-1}{(2n+6) \cdot 6} \left(\frac{a_0}{2! 2^4 (n+1)(n+2)} \right)$$

$$\boxed{a_6 = \frac{-a_0}{2^6 3! (n+1)(n+2)(n+3)}}$$

Now By eqn (2)

$$y = \sum_{k=0}^{\infty} a_k x^k = x \sum_{k=0}^{\infty} a_k x^k$$

Here the case continue for $k=n$, so

$$y = x^n [a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots]$$

$$y = x^n \left[a_0 - 0 + x^2 \left(\frac{-a_0}{2^2(n+1)} \right) + 0 \right.$$

$$+ \left(\frac{a_0}{2! 2^4(n+1)(n+2)} \right) x^4 + 0 + \left(\frac{-a_0}{2^6 3!(n+1)(n+2)(n+3)} x^6 \right. \\ \left. + \dots \right]$$

$$y = a_0 x^n \left[1 - \frac{x^2}{2^2(n+1)} + \frac{x^4}{2! 2^4(n+1)(n+2)} \right. \\ \left. - \frac{x^6}{2^6 3!(n+1)(n+2)(n+3)} + \dots \right]$$

$$y = a_0 x^n \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{2^k k! (n+1)(n+2) \dots (n+k)}$$

$$\text{Put } a_0 = \frac{1}{2^n \Gamma(n+1)} \left(= \frac{1}{2^n} \frac{1}{n!} \right)$$

$\because n! = \Gamma(n+1)$

$$= \frac{1}{2^n \Gamma(n+1)} x^n \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{2^k k! (n+1)(n+2) \dots (n+k)}$$

$$= \sum_{l=0}^{\infty} \frac{(-1)^l x^{n+2l}}{2^{n+2l} l! n! (n+1)(n+2)\dots(n+l)}$$

$$y = \sum_{l=0}^{\infty} \frac{(-1)^l \left(\frac{x}{2}\right)^{n+2l}}{l! \Gamma(n+2l+1)}$$

It is Bessel's function of first kind of order n and denoted by

$$J_n(x).$$

$$J_n(x) = \sum_{l=0}^{\infty} \frac{(-1)^l}{l! \Gamma(n+2l+1)} \left(\frac{x}{2}\right)^{n+2l} \quad \text{--- } \textcircled{\#}$$

Case II For $k = -n$

$\textcircled{\#} \Rightarrow$

$$J_{-n}(x) = \sum_{l=0}^{\infty} \frac{(-1)^l}{l! \Gamma(-n+2l+1)} \left(\frac{x}{2}\right)^{-n+2l}$$

General solution of Bessel's eqn.

$$y = A J_n(x) + B J_{-n}(x) \quad \text{--- } \textcircled{\#}$$